

Danniell Hu

(734) 747-0278 | dannihu@umich.edu | dannihu01.github.io

EDUCATION

University of Michigan, Rackham Graduate School

Ph.D., Computer Science, GPA 4.0/4.0

Ann Arbor, MI
Expected Apr. 2029

Advisors: Dr. Elizabeth Bondi-Kelly, Dr. Westley Weimer

University of Michigan, College of Engineering

B.S.E., Computer Science, Cum Laude

Ann Arbor, MI
May 2023

RESEARCH EXPERIENCE

University of Michigan

Ph.D. Researcher

Ann Arbor, MI
2024 – Present

- Research examines the two-way relationship between AI systems and human context: how stakeholder needs, constraints, and values shape AI development, and how deployed AI reshapes users' decisions, skills, and capacities — grounded in real-life, high stakes health settings.
- Conducted a controlled human-subjects study (n=49), providing empirical results showing that the ability to reason about human goals, values, needs, and constraints correlates to AI system designs more aligned to stakeholders, not traditional requirements engineering metrics or prior theoretical or practical AI experience (AIES 2026).
- Developing PATHFinder, an LLM-powered prenatal care planning system, and leading mixed-methods user research with patients, clinicians, and community health workers to examine trust, transparency, and clinician oversight in multi-stakeholder clinical AI; targeting CHI 2026.
- Published a human-AI collaboration framework in AI Magazine (2025) synthesizing design considerations across fairness, labor, privacy, and human agency.

University of Michigan

Undergraduate Researcher

Ann Arbor, MI
2022 – 2024

- Conducted a neuroimaging study of n=28 programmers using functional near-infrared spectroscopy (fNIRS) to investigate the cognitive processes underlying end-to-end debugging.
- Proposed and validated a five-stage cognitive model of debugging (Task Comprehension, Fault Localization, Code Editing, Compiling, Output Comprehension), providing the first neurological evidence that these stages are both behaviorally and neurally distinct (IEEE TSE, 2024).
- Designed and analyzed controlled identifier morphology conditions to study how variable naming impacts cognitive load for programmers with reading difficulties (dyslexia, non-native English speakers).

INDUSTRY EXPERIENCE

Stryker Corporation – Medical Division, Acute Care

R&D Embedded Software Design Engineer

Kalamazoo, MI
July 2023 – Aug 2024

- Designed embedded software and PCBA hardware for Stryker medical devices, including hospital beds and support surfaces, using ColdFire and Renesas microcontrollers.
- Collaborated with cross-functional teams across test lab, quality assurance, hardware, electrical, software, marketing, and leadership.
- Participated in customer visits contributing insights that led to multi-million dollar sales opportunities.

Stryker Corporation – Medical Division, Acute Care

R&D Embedded Software Design Intern

Portage, MI

Summers, July 2020 – July 2022

- Automated test suites for large-scale, multi-medium embedded software systems using Jenkins and Artifactory, improving CI/CD pipeline efficiency.
- Researched and prototyped a high-level real-time tracking system integrating UWB, Bluetooth, and infrared technologies for existing embedded product lines.
- Developed computer vision and machine learning models to improve patient safety around hospital beds, including patient tracking, pose detection, object recognition, and image obfuscation.

PUBLICATIONS & PREPRINTS

- **Hu, Danniell**, Westley Weimer, and Elizabeth Bondi-Kelly. "Towards Stakeholder-Grounded Requirements Engineering in AI System Scoping." *AI, Ethics, and Society (AIES) 2026*
- **Hu, Danniell**, Diana Acosta Navas, Susanne Gaube, Hussein Mozannar, Matthew E. Taylor, Krishnamurthy Dvijotham, and Elizabeth Bondi-Kelly. "Human at the Center: A Framework for Human-Driven AI Development." *AI Magazine*, 46(4), 2025. DOI: 10.1002/aaai.70043
- **Hu, Danniell**, Priscila Santiesteban, Madeline Endres, and Westley Weimer. "Towards a Cognitive Model of Dynamic Debugging: Does Identifier Construction Matter?" *IEEE Transactions on Software Engineering*, 50(11), 2024. DOI: 10.1109/TSE.2024.3465222

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Javascript, etc.]

ML / AI: PyTorch, [Hugging Face, scikit-learn, etc.]

Tools: Git, LaTeX, [Docker, etc.]

Research: Requirements Engineering, Participatory Design, User Studies, Statistical Analysis

HONORS & AWARDS

e-HAIL Dataset Creation Award

2026

University of Michigan E-Health & Artificial Intelligence — \$5,000 award.

KLA Fellowship

2024

University of Michigan Rackham Graduate School — \$45,000 fellowship award.

Ford Blue Oval STEAM Scholarship

2019 – 2023

Ford Motor Company — \$10,000 annually.

- Invitation-only program that awards scholarships to college-bound students who have participated in select Ford STEAM programs and meet selection criteria. Recurring annually upon academic standing.

SERVICE & LEADERSHIP

University of Michigan, CSE Graduate Student Organization (CSEG)

President

Ann Arbor, MI

May 2026 – Present

- Elected to lead CSEG's executive board, setting the organization's strategic direction, budget priorities, and programming on behalf of the department's entire graduate student body.
- Develop the leadership capacity of a 14-officer board through one-on-one mentorship, delegated ownership of initiatives, and structured role transitions.
- Coach first-time officers to run programs independently and grow into future board leadership.
- Represent graduate student interests directly to CSE department leadership, advocating for student priorities in faculty and administrative decision-making.
- Cultivate a connected, inclusive department community through active and thoughtful outreach, engaging students across cohorts, labs, and degree programs.

University of Michigan, CSE Graduate Student Organization (CSEG)

Social Chair

Ann Arbor, MI

Aug 2025 – May 2026

- Plan and execute recurring department-wide social events (e.g., weekly Tea Time, seasonal mixers, game nights, off-campus outings) to strengthen graduate student community and faculty engagement.
- Build ongoing feedback mechanisms with CSEG members and use qualitative and quantitative input to refine programming and improve responsiveness to student needs.
- Advocated for and led the redesign of CSEG website to improve accessibility of resources and initiatives.
- Facilitated team-building activities, cohort cohesiveness, and inter-departmental collaboration.

University of Michigan, AI Lab

AI Blog Editor

Ann Arbor, MI

Aug 2025 – August 2026

- Write, edit, proofread, and publish articles for the University of Michigan AI Lab blog, maintaining a consistent publication cadence and ensuring editorial quality.
- Recruit and support graduate student contributors to broaden research communication across the lab community.

MENTORSHIP & ADVISING

Jett Rosen

SUGS M.S. Student, Computer Science — Research Mentee

May 2026 - Present

- Supervise the student's first-authored research study of AI scribes in a pro-bono physical therapy clinic, guiding them from research question formulation through study design, data analysis, and manuscript writing; targeting NeurIPS workshop 2027
- Meet weekly to set research milestones, review analyses and drafts, and provide structured feedback to build the student's independent skills in study design, qualitative methods, and academic writing.

REFERENCES

Available upon request.